

Supplementary Material for Political Polarization and Fractionalisation from Rational Values-Based Inference in an Agent-Based Graph Network

1 Graph tools

Here we present some of the technical details necessary to reproduce the results in the main paper. All code is available on the anonymous OSF repository https://osf.io/a249r/?view_only=ae9df0db778141a4b224c3ef7826b2ad.

1.1 Graph normalization

The graphs G_a and G_b are generated by the KL-divergence of moral representations represent a distance between nodes. To conduct spectral clustering analyses we convert these distance graphs into similarity graphs. Given that the distance between any two nodes is strictly positive we normalize the similarity by the following equation:

$$s_{i, a, t} = e^{-b \cdot w_{i, a, t}} \quad (1)$$

where b is a normalization constant that we fit to maximize the variance of the weights in the entire network over an entire simulation.

1.2 Spectral clustering

Many of the results in the main paper depended on finding clusters in the network at different points in time. Here we detail the steps to computing the optimal number of clusters given the normalized similarity matrix as defined in the previous section.

We simplify the bi-directional similarity matrix by taking the minimum similarity between any two nodes, and thus rendering the matrix symmetrical. We then compute the Eigen-values of the Laplacian of the simplified similarity matrix. Following methodology in von Luxburg, 2007, the optimal number of clusters is given by the largest gap in Eigen-values. We restrict to clusters of size 2 or larger so that meaningful clusters are formed.

2 Alternative model implementations

2.1 Moral Value Normalization

While the beta-binomial update of moral values is a convenient feature that follows from representing moral values with a beta distribution, another result is that the moral values become increasingly more confident over time—lowering the effect that evidence has on updating changing values later in the simulation. To mitigate this, we consider belief update where the distribution parameters are always normalized such that the resulting $\alpha_{norm} = \frac{\alpha}{\alpha+\beta}$ and $\beta_{norm} = \frac{\beta}{\alpha+\beta}$.

We found that these initializations had similar effects on the general results of our simulations (cf. figures below).

References

von Luxburg, U. (2007). A tutorial on spectral clustering. *Statistics and Computing*, 17(4), 395–416. <https://doi.org/10.1007/s11222-007-9033-z>

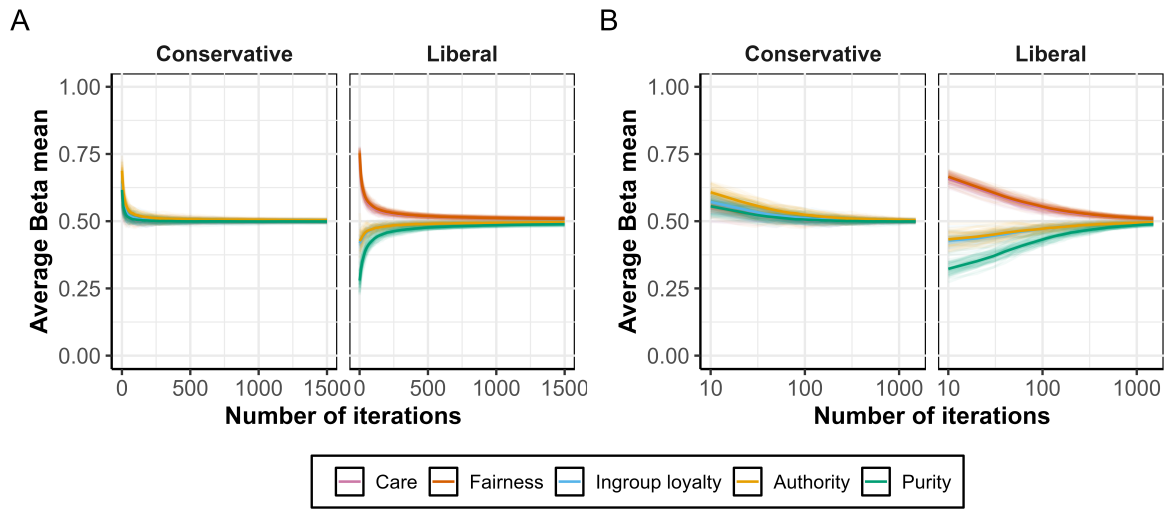


Figure 1: Normalized model implementation: Mean agent moral values by identity over (A) time and (B) log(time). Thick lines indicate the trend across simulations, and faint lines represent individual simulations.

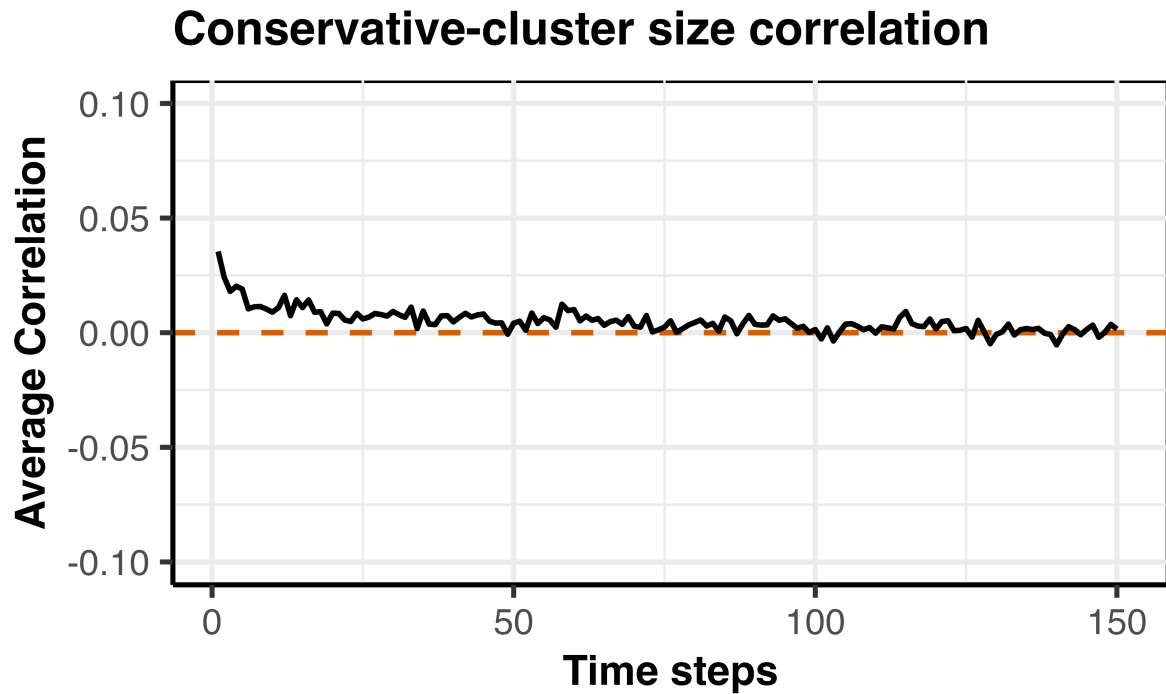


Figure 2: Normalized model: (A) Correlation between cluster size and proportion of conservatives over time averaged over simulations. (B) Example time step showing how correlations were derived. The blue x in (A) indicates the time step shown in (B).

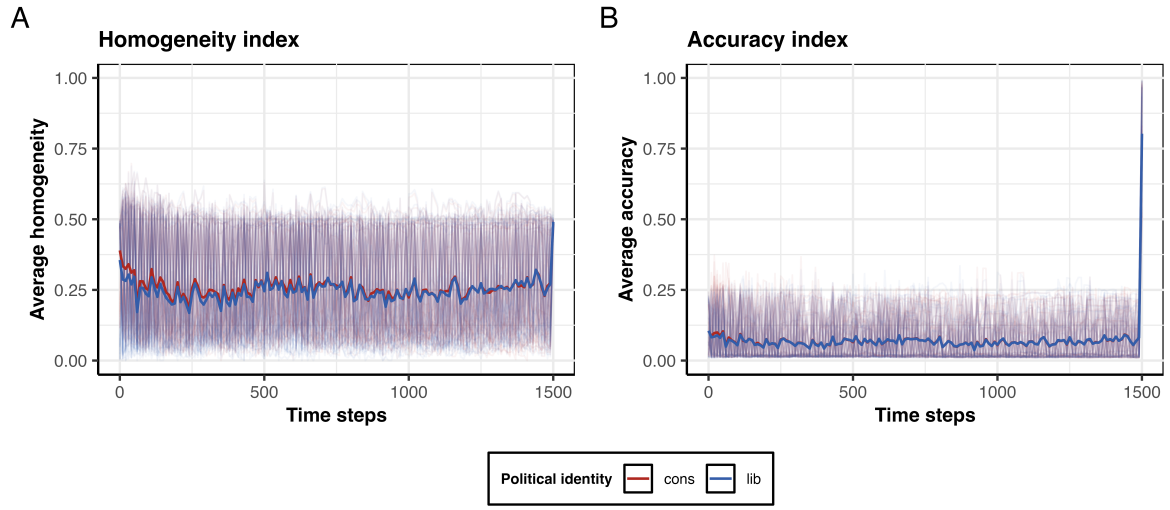


Figure 3: Normalized model implementation: Average (A) homogeneity and (B) accuracy over all simulations (thick lines) and individual simulations (faint lines).

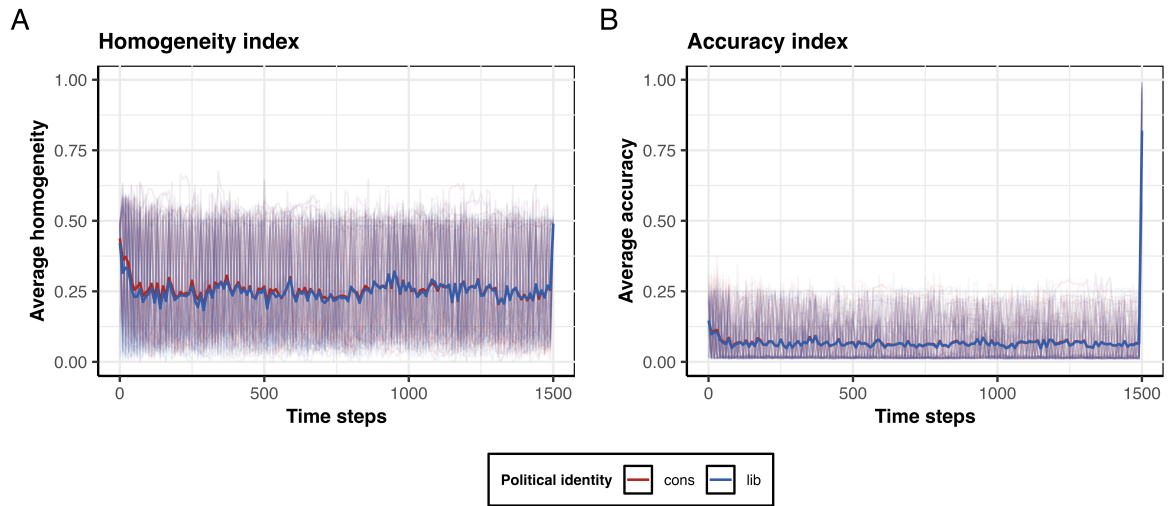


Figure 4: Main model: Average (A) homogeneity and (B) accuracy over all simulations (thick lines) and individual simulations (faint lines).